

# Arun Venkatachalam

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## Skills

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- Python for Data Science, Data Analytics Packages, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn, Statistics, EDA, Hypothesis testing, SQL Advanced, PySpark
- Machine Learning – Regression, Classification, Clustering, Recommendation Systems, Gradient Boosting (XGBoost), SVM, k-NN, Naïve Bayes
- Deep Learning Techniques - CNN, RNN, LSTM, GAN, diffusion models, Transformers, Image synthesis, Reinforcement Learning
- Version Control - Git and Github
- Natural Language Processing - NLTK, Spacy, RegEx, HuggingFace, Speech Recognition, Gen AI, Large Language Models and Fine Tuning
- Computer Vision - OpenCv, Tensorflow, Keras, YOLO, RCNN, PyTorch, CUDA architecture, C++ Programming
- Docker, Kubernetes, Flask, Fast APIs, Streamlit, Langchain, Linux Machine, Applied Mathematics for ML & DL
- MLOps & Cloud - AWS Cloud Services, Google Cloud Platform, Microsoft Azure, Airflow, ML Flow
- Strong Communication and Interpersonal Skills, Analytical and Problem-solving skills, Technical Presentation, AI solutions, Workflow Automation, Model Training, Production deployment responsibility
- ML Solution Deployment – Deploying and maintaining scalable, efficient, and production-ready ML pipelines using containerization, orchestration tools, and cloud infrastructure

## Experience

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**ML Engineer, Positive Integers Pvt Ltd – Chennai, IN** Feb 2025 – Present

- Designed and deployed production-grade Generative AI and machine learning solutions integrated into enterprise client workflows using PyTorch and LangChain
- Implemented scalable ML pipelines using Kubernetes, Docker, and Airflow supporting distributed training and batch inference workloads
- Developed recommender system-based automation modules for FMCG supply-chain optimization using historical demand forecasting models
- Built CI/CD-enabled ML deployment pipelines across AWS, GCP, and Azure ensuring scalable and maintainable model lifecycle management
- Optimized inference latency and pipeline efficiency through containerized ML services deployed in cloud-native environments
- Collaborated with cross-functional teams to productionize ML algorithms including regression, classification, and recommendation engines
- **Tools & Technologies:** Gen AI, LLM models, LLMops, Langchain, Kubernetes, Docker, NLP, ML models, Deep Learning models, PyTorch, Kubernetes, Docker, Airflow, Regression, Classification and Recommendation algorithms

**Data Scientist - External Consultant, Positive Integers Pvt Ltd – Chennai, IN** Aug 2024 – Jan 2025

- **Automated Suggested Order Module:** Developed an automated order processing system for FMCGs based on historical data using MLOps practices.
- **Optimized Workflow:** Automated, scheduled, and monitored the entire workflow using Airflow UI, resulting in a 5% reduction in costs and a 40% reduction in processing time compared to dataproc clusters.
- **CI/CD Pipeline:** Implemented a CI/CD pipeline with tools such as GitHub Actions, Airflow on Kubernetes, Artifact Container Registry, Snowflake, and Google Cloud Storage (GCS).
- **Tools & Technologies:** SQL, Kubernetes, Docker, Airflow, Snowflake, XGBoost, and VS Code

**Data Scientist, Positive Integers Pvt Ltd – Chennai, IN**

July 2023 – May 2024

- Analyzed & addressed a challenge at a popular FMCG company where 3.2% of promoter outlet sales were allocated towards commissions, with a focus on reducing this percentage
- Derived outlet-level target distribution based on historical sales trends data analysis and developed a robust **Suggested Order Module (SOM)** for FMCGs to streamline order processing for various SKUs
- Utilized AWS SageMaker instance (Linux) to streamline SOM processes by extracting historical data from Redshift database, Performed in-depth analysis of data to uncover underlying characteristics & patterns
- Tools used: AWS Sagemaker, AWS Lambda, AWS Secret Manager, AWS Redshift, Python

**Data Science Intern, Positive Integers Pvt Ltd – Chennai, IN**

June 2023 – Jul 2023

- Leveraged Transformer pre-trained LLMs to develop a proprietary GPT model tailored for generating personalized recommendations in a business setting
- Utilized Direct Connect to enhance data security & expedite processing in AWS EC2 instance-to-instance data transfers
- New Feature creation for FMCG business data, Evaluating the impact of data-driven projects and initiatives on the organization's bottom line
- Converted PySpark to Python code for data science models, implementing machine learning algorithms to transition regression & classification models

## Projects

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### Generative AI Projects

- Developed and implemented various Gen AI projects, including an ATS resume tracker, YouTube summarizer, Q&A chatbot, Nutritionist Gen AI Doctor, and PDF document query system, aimed at enhancing everyday tasks and providing intelligent automation solutions
- Tools Used: Langchain, Streamlit, Pinecone Vector DB, Open Source LLMs (LLAMA, MISTRALmodels), Paid Source LLMs (OPENAI LLM), FAISS, GROQ Inference Engine(LPU), AWS Sagemaker, AWS Bedrock, AWS Lambda, AWS API Gateway
- Link : [github.com/aruneer007/Generative-AI](https://github.com/aruneer007/Generative-AI)

### Water Quality Classification

- Since freshwater is one of our most vital and scarce natural resources, making up just 3% of the earth's total water volume, this project aims to Predict whether fresh water from various sources and with chemical parameters is safer to drink
- Tools Used: Python, VS code, Scikit-learn, Flask, AWS Elastic Beanstalk, AWS Code Pipeline
- Link : [github.com/aruneer007/WaterQualityClassification](https://github.com/aruneer007/WaterQualityClassification)

### Book Recommender System

- Developed collaborative filtering and content-based recommender systems and deployed as production-ready API services using Flask and AWS Elastic Beanstalk
- Tools Used: Python, Recommendation Systems, VS Code, Flask API, AWS Elastic Bean Stalk, AWS Code Pipeline
- Link : [github.com/aruneer007/Bookrecommendersystem](https://github.com/aruneer007/Bookrecommendersystem)

## Education

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**Kongu Engineering College**, BE in Electrical and Electronics

Jul 2011 – May 2015

- **Coursework:** Digital Electronics, Analog Circuits, Digital Signal Processing, Power Electronics, Microprocessor and Microcontrollers, Computer Architecture

**Great Lakes Institute of Management**, PG Program in Artificial Intelligence and Machine Learning

Dec 2021 – Jan 2023

- **Coursework:** Statistics, Supervised and Unsupervised Learning, Recommendation Systems, Deep Learning, Computer Vision, Natural Language Processing

## Technologies

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**Languages:** C++, Python, SQL

**Technologies:** Linux, VS Code, AWS Cloud Services, Google Cloud Platform, Docker

## Certificates

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**Post Graduate Program in Artificial Intelligence and Machine Learning** Jan 2023

- Link : [Credential URL](#)

**SQL Beginner to Advanced For Data Professionals** Jan 2023

- Link : [Credential URL](#)